

Investigating Interaction Modes and User Agency in Human-LLM Collaboration for Domain-Specific Data Analysis

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ABSTRACT

Despite demonstrating robust capabilities in performing tasks related to general-domain data-operation tasks, Large Language Models (LLMs) may exhibit shortcomings when applied to domain-specific tasks. We consider the design of domain-specific AI-powered data analysis tools from two dimensions: **interaction** and **user agency**. We implemented two design probes that fall on the two ends of the two dimensions: an open-ended high agency (OHA) prototype and a structured low agency (SLA) prototype. We conducted an interview study with nine data scientists to investigate (1) how users perceived the LLM outputs for data analysis assistance, and (2) how the two design probes, OHA and SLA, affected user behavior, performance, and perceptions. Our study revealed insights regarding participants' interactions with LLMs, how they perceived the results, and their desire for explainability concerning LLM outputs, along with a noted need for collaboration with other users, and how they envisioned the utility of LLMs in their workflow.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Large language model, data analysis, domain knowledge, human-AI collaboration, user agency

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1 INTRODUCTION

Domain-specific data operations encompass a range of activities, including data transformation, processing, and analysis across specialized domains, such as medicine, manufacturing, finance, sports, and beyond. Conducting complex data operations requires a deep understanding of the specific data structures and thorough knowledge of the unique concepts and terminologies relevant to each domain.

In recent years, pre-trained Large Language Models (LLMs) have demonstrated significant capabilities in diverse data science and machine learning tasks, such as visualization [26], junior-level data analysis [5], classification [12], and model selection [22]. Many research communities and companies started to investigate Human-LLM collaboration as the future of programming, as studies showed that LLMs can save developers' searching efforts [24], improve productivity and developer happiness [14]. Commercially available tools such as Github *Copilot* and ChatGPT *Code Interpreter* [19] garnered significant attention upon their release.

Despite demonstrating robust capabilities in general-domain data operations, LLMs may exhibit shortcomings when applied to domain-specific tasks. While the extensive training data enables LLMs to comprehend a wide range of concepts or terms, it may still be insufficient for understanding meanings specific to certain organizations or nuanced contexts.

We consider the design of AI-powered domain-specific data analysis from two dimensions: **interaction** and **user agency**, as shown in Table 1. Many conversational LLM applications offer **open-ended interactions**, requiring users to consider their prompts to LLMs, with the freedom to use code, natural language commands, or questions. **Structured interaction** typically features predefined stages and ensures that the user's journey through a system follows a logical path. Users are guided by the system to complete certain steps to get results. **User agency** or human control is a critical aspect in automated systems [11, 23]. Code assistants such as Github *Copilot* offer suggestions, leaving it up to the user to decide which code snippet to use, depending on the user's actions like code execution to achieve the result. In contrast, LLM agents such as ChatGPT *Code Interpreter* execute the code on behalf of the user and automatically present the results. AI-powered tools for data analysis such as PandasAI [25] and VizGPT [1] provided automatic workflow to convert user queries into visualization or analysis results. Besides user agency on the **code execution** level,

Interaction	User agency		Application
	Execution	Planning	
Open-ended	high	high	Copilot, ChatGPT, * Design Probe OHA
Open-ended	low	high	Code Interpreter, PandasAI, VizGPT
Structured	low	low	* Design Probe SLA

Table 1: AI-powered coding tools on interaction and user agency dimension. We implemented two design probes on the ends of the two dimensions: an open-ended high agency (OHA) that has an open-ended workflow and gives users high execution and planning agency on LLM outputs; another is a structured low agency (SLA) application that guides users step-by-step and automatically plans and executes code to complete tasks.

planning-level assistance is also a critical aspect in data analysis and has also been discussed [9]. However, little work has explored structured interaction in AI-powered data analysis tools. In the context of **domain-specific** tasks, human-LLM collaboration is essential as LLMs often lack sufficient domain knowledge.

In this work, we explore the design space of domain-specific data analysis by inviting users to work with two probes from the ends of the two dimensions: one has an open-ended interaction workflow and gives users high agency on both execution and planning level; the other has a structured interaction workflow that guides the user step-by-step and gives lower execution and planning agency compared with the first one. We conducted an interview study with nine participants and invited them to complete two domain-specific analysis tasks on each prototype.

We aim to answer the following research questions:

- **RQ1:** How do users perceive the outputs of LLMs in the context of domain-specific data analysis tasks?
- **RQ2:** What is the optimal design for LLM-powered tools to assist in data analysis tasks?

Our key findings are:

- Participants expressed overall satisfaction with the LLMs in assisting them with data analysis tasks, while raising concerns about the need for transparency and ability to verify results.
- The different interaction modes (open-ended and structured workflow) and the two degrees of agency (high and low) resulted in different user behaviors, each with its own set of advantages and limitations.

Based on these findings, we propose design implications for designing an LLM-enhanced collaborative system for supporting data analysis tasks.

2 INTERVIEW STUDY

2.1 Design Probes

We implemented two interactive prototypes instead of using slide-based prototypes [20] or Wizard-of-oz method [9] because we want the users to experience the application as close to their daily job as possible. We opted not to employ existing tools such as Github Copilot or ChatGPT because we want the two prototypes to use the same LLM to make the output consistent. *Code Interpreter* is a commercially available tool but its data analysis requires users to upload data files, which is not applicable for enterprises that concern data security and sensitivity.

OHA (Figure 1-A) is an LLM-based chat web application integrated with a rudimentary API that connects to Azure OpenAI GPT-4 service. As a vanilla chat interface, it is limited to basic messaging functionalities. When used in data analysis tasks, users still need a developer environment to execute code.

SLA (Figure 1-B) is a structured, low-agency LLM-powered application. It guides users through a data analysis workflow, makes plans, and executes the code on behalf of the user. We designed the workflow based on literature about data analysis [16, 28] and LLM prompt engineering [27, 29]. The workflow is as follows:

- (1) Describe data (User). Users can upload multiple CSV files, edit descriptions of each file, and provide context information about the analysis, such as data source and domain (see Figure 3 in Appendix). LLMs tend to perform better with more contextual information. Dibia [7] examined that providing enriched data summary leads to reduced error rate when generating visualizations.
- (2) Inject query and domain knowledge (User). Users can inject a question they want to investigate and domain knowledge that may be helpful for the LLM to complete the task.
- (3) Enhance prompt (System). The system generates a step-by-step guide to complete the task based on the question and domain knowledge the user provided (see Figure 4 in Appendix).
- (4) Select enhanced prompt (User). Users can select from three variations, then the user can trigger execution.
- (5) Automatic data analysis (System). This step involves code generation, execution, debugging, and insight generation. The backend is an LLM agent that adapts the ReAct framework [10, 29], which orients the LLM to iterate through reasoning, action, and observation cycles.
- (6) Review output (User). Users can view the result of execution, including visualizations, code, and insights generated by SLA (see Figure 5 in Appendix).

2.2 Tasks and Procedure

Following a lab experiment study guideline, we conducted a within-subject study to compare the interactive system built upon our framework and existing LLM applications.

Participants. We recruited nine data scientists from a product team in a multi-national enterprise (8M, 1F), aged from 25 to 46). Their job content includes data analysis in manufacturing and sensor domains. All of them have at least one year of working experience in data analysis. Participants self-identified job titles are

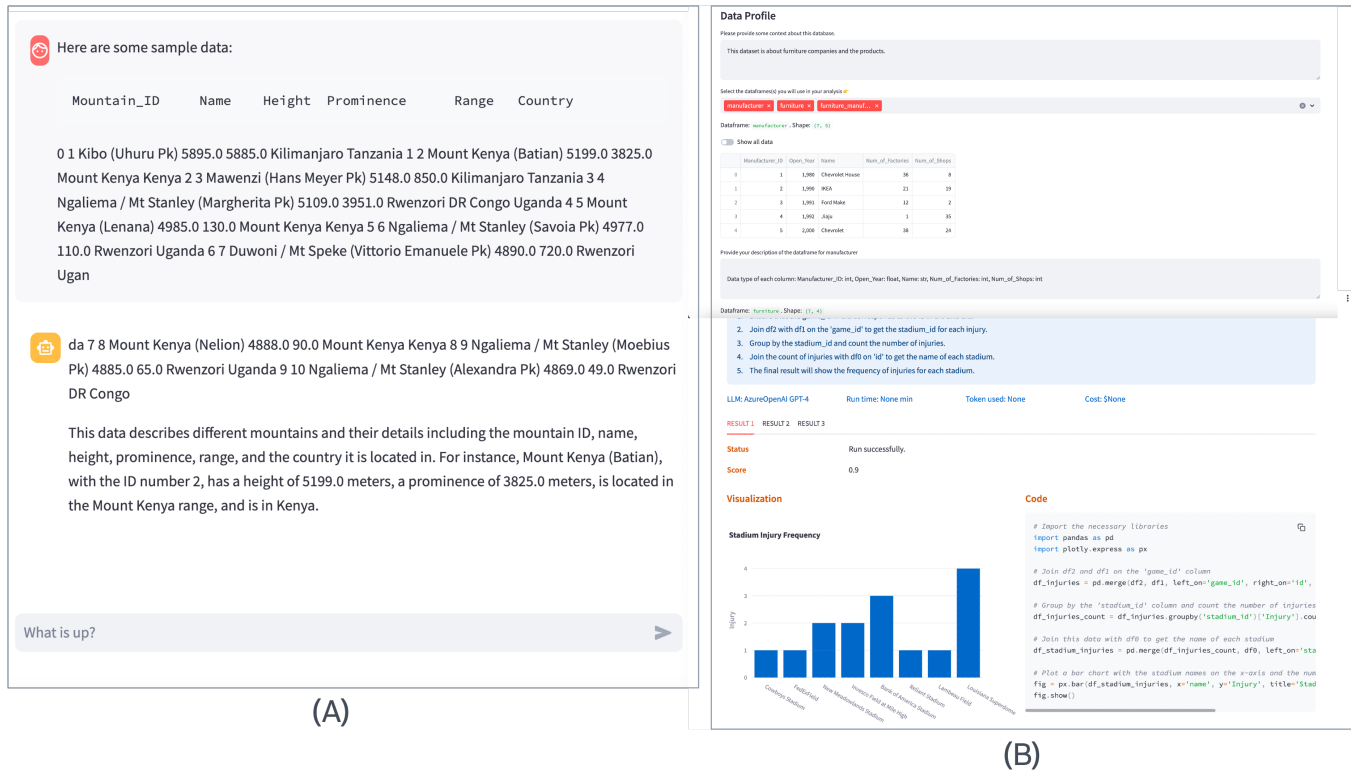


Figure 1: A: Interface of design probe OHA. B: Interface of design probe SLA.

data scientist (6), data engineer (1), and research scientist (2). Participants have various degrees of familiarity with LLM applications for data analysis. Some participants have experience of using LLMs to polish writing and summarize articles but have not used LLMs in data science job. Some express distrust towards LLMs, and use LLMs to generate code templates or overall structures only. Two use LLMs frequently in their daily work. Each participant received \$20 gift card as compensation.

Before the study, participants completed a pre-study survey asking for demographic information and their previous experience with LLM-powered applications. The interview study can be broken down into three parts.

Part 1: Pre-study interview. We first asked some questions about participants' job content that is relevant to data analysis. Participants were cued to recall an experience when they needed to deliver data analysis reports to stakeholders, the volume of data to be analyzed, the negotiation procedure, and the format of the report.

Part 2: Two domain-specific data analysis tasks. Participants were asked to complete two data analysis tasks on enterprise internal data with the help of LLM-powered tools, one is OHA, and the other is SLA. Participants were encouraged to "think aloud" while they were doing the task. The order of tools was randomized. We want to simulate participants' daily work as much as possible, therefore the data used in the study were extracted from a huge

data lake to which the authors have access. The two tasks used the same data source but different requirements (see Figure 2).

Part 3: Critiques and reflections. After two tasks, participants critiqued and reflected on their experience by comparing the two tools, and discussing the perceived utility and limitations.

Interviews were conducted by two researchers. One researcher led the interview and the other took notes. The two researchers had retrospective meetings after each interview and discussed the takeaways. All the interviews were video-recorded and with participants' permission and transcribed. Using an inductive open-coding analysis approach [6, 21], the authors read through the transcripts and watched recordings multiple times to conduct the initial coding. Then, both authors read through and discussed the coded transcripts together, iteratively highlighted excerpts, and identified key insights and recurring patterns in the data.

3 PRELIMINARY FINDINGS

In this section, we discuss the insights from our probe interviews, which we have organized according to our two main research questions: (1) how users perceived the LLM outputs for data analysis assistance, and (2) how the two design probes, OHA and SLA, affected user behavior, performance, and perceptions.

Prompt LLM tools to answer the question. Output: python code, visualization (if any), insights

Question 1

In Plant A, does the current station contribute to next station's anomaly?

Question 2

In Plant B, is there any evidence that previous station affect the next station's anomaly?

Figure 2: Example tasks in the interview study. Participants received the example task as a prompt and were informed that they could customize their own queries that fit their job.

3.1 User Perception of LLM outputs for Domain-specific Data Analysis

In terms of how users perceived LLM outputs for their data analysis tasks across the two design probes, we identified three broad themes:

3.1.1 Output Quality. Participants appreciated OHA for its ability to assist in quickly generating initial versions of code, which was seen as a valuable feature for starting data analysis tasks, even though executing the code threw errors at times. P5 said, *"It's quite easy and intuitive, especially for easy tasks, they are just supposed to write out some code."* They also expressed similar satisfaction with the insights produced by SLA, acknowledging its utility in enhancing workflow efficiency, especially during the exploratory phase and reporting stage. The prototype's ability to generate comprehensive plots and explanations was particularly valued for its potential to aid communication with stakeholders.

I think for the first time even for us to think what to tell to customer from this graph, it takes a little bit time to think how to explain. But here, if we read these insights, it gives some tips what can make sense... To start with, it is really good. Maybe a little bit more articulation or additional information might be required when we are telling to the user. I think this insights are helpful. (P8)

Participants expressed overall satisfaction with the code and insights produced by LLMs across both probes, even though some conclusions lacked in-depth findings or did not align with the problem. Despite these limitations, participants were optimistic about how the LLM outputs here can be a promising starting point for future iterations.

3.1.2 Trust and Verifiability. Trust emerged as a crucial concern among participants using SLA, particularly due to challenges in verifying the tool's results. Users expressed hesitation in relying on the system without independent verification, underscoring the need for trust-building measures. P2 encapsulated a similar sentiment for OHA: *"I'm seeing how nice that was. It was really quick to get an answer. I don't know how to trust it without actually redoing all the work myself, which would be very time-consuming."* The lack of domain-specific fine-tuning in LLMs further contributed to this trust issue, with users like P1 and P4 voicing concerns about the system's understanding of their intentions and domain knowledge.

Additionally, the inconsistency in the tool's outputs, attributed to the LLM's *temperature* setting, was found to impact trust, especially for users less familiar with data analysis methodologies.

3.1.3 Need for Explanations. Participants emphasized the importance of explanations and transparency when interacting with LLMs. They actively sought clarity on the LLM's reasoning process, particularly when outputs did not match expectations. P2 captured this need: *"Cause at least it gives me an idea that it understands the concept and what to do... I could edit this if I was doing a real thing and make sure that this is exactly the right steps."* This desire for explanations was especially pronounced in situations where the LLM suggested complex operations, like machine learning models, but did not follow through, leading to dissatisfaction and confusion. P7's experience underscores this frustration: *"It completely over promised the delivery of the machine learning model... I'm very unsatisfied that it does not give me a model at all and also does not explain to me why."* This finding highlights a need for more transparent communication of the LLM's decision-making process to build user trust and understanding.

3.2 Impact of Design Modes on User Behavior

Here, we discuss how the interaction modes (i.e., open-ended and structured) and the user agency (i.e., high v/s low) in the two probes affected user behavior and perception of the LLM prototypes.

3.2.1 Interaction Modes. We examine the impact of the two distinct user workflows — open-ended (OHA) and structured (SLA) — on user behaviors (see Table 2) and perceptions. In the open-ended OHA mode, participants with LLM experience often included sample data or key columns in their queries, seeking to provide context. P4 highlighted the cumbersome nature of this process: *"I wish we have a better way to directly inject this code and meta informational data into this system... I have to manually write down this prompt by myself, which is somewhat not easy."* Conversely, in the structured SLA mode, users typically did not engage in an interactive dialogue about the data. Instead, they provided brief text descriptions or essential details about the dataset, relying on the system's structure to guide the data analysis process.

For domain knowledge, OHA users often embedded this information within their questions or as side notes, while SLA users provided varying levels of detail, intuitively gauging what the model might need. P1's experience illustrates this: *"With the domain knowledge, I can be more clear about what features and values I want... Like traffic lights on the left, maybe the language model would interpret this as high values for left, but instead, this is low values for left."* This varied approach to input in SLA sometimes led to users selecting the first provided LLM-enhanced domain knowledge variation and proceeding without exploring other options, citing time constraints. This structured approach in SLA helped guide users in providing necessary information, sometimes filling in gaps that were more challenging in the open-ended OHA mode.

3.2.2 Agency. In the OHA setup, users demonstrated high agency by independently determining the course of their analysis, making decisions without system guidance. In contrast, the SLA probe guided users through the process, providing detailed domain knowledge and outlining analysis steps, thus reducing the level of user

agency in planning. Error handling in OHA demonstrated high user agency, with participants actively diagnosing and communicating errors to the LLM. Users copied the entire or partial error messages, directly engaging in the debugging process. In contrast, the SLA probe exhibited low user agency in error handling; most errors were automatically managed within the workflow, often remaining unseen by users.

However, in cases of unresolved errors due to token limitations, user agency was limited to rephrasing inputs without direct involvement in error resolution. Participants across both setups underscored the need for iteration capabilities in LLM-powered tools: *"I think it does need some iteration with the model, and then after you take the output of the model, you need to adapt it a little bit on your own."*

4 DISCUSSION

The design probes used in the interview were designed and implemented to align with the two ends on the **interaction** and **user agency** dimensions. In this section, we discuss the findings and design considerations from these dimensions. Then we discuss the trust and verifiability of LLM-powered data analysis tools and design recommendations.

4.1 Design Considerations of AI Assistance for Domain-specific Data Analysis

Structured Interaction. In HCI literature, structured workflows have been widely adopted in interactive applications. This approach, which predates the advent of LLM applications such as ChatGPT, is designed to streamline user workflows and guide users to prevent errors. Since the launch of ChatGPT, conversational and open-ended interaction have grown in popularity, offering more flexible and natural interaction paradigms. Many AI-powered tools for data analysis [1, 25] and research-oriented design prototypes have adopted the open-ended interaction paradigm [9, 20]. Despite allowing users the freedom to initiate conversations, open-ended interaction necessitates that users possess prompt engineering skills, such as choice of terminology, arrangement of word order, as well as repetition in keywords [30]. This requirement may burden data analysts with the additional task of mastering prompt engineering if their original goal of using AI-powered tools is to increase efficiency [3, 20]. For users inexperienced with prompt engineering, predefined steps aid in understanding how LLMs function. Intermediate steps elucidate how the LLM interprets tasks and decides on actions, enhancing transparency. The appropriateness of conversational interaction for LLM-powered data analysis remains an open question. Designers and researchers should avoid creating chatbots merely for their own sake. Instead, considering a system that incorporates both open-ended and structured interactions may be beneficial. For instance, developers could implement a conversational assistant that guides users through predefined steps, or a structured interface that allows for configurable steps.

Execution-level User Agency. Participants expressed appreciation for the automated processes in code generation, debugging, and interpreting results. This finding suggests that low levels of control at the execution level do not result in dissatisfaction among

our participants. It was suggested that increased agency, allowing data analysts to modify the code before and after execution, would be beneficial. We also found that depending on the job content, the range of execution-level tasks may be expanded to data wrangling pipeline instead of only code generation and execution. Participants expressed a **need for automation in foundational data pre-processing and wrangling tasks**, such as normalizing data, filling in missing values, extracting descriptive statistics, finding outliers, and visualizing distribution. These tasks, despite their repetitive nature, are crucial for thorough analysis [8]. Our observations confirmed that participants consistently prioritize these initial steps, underscoring the potential benefits of streamlining them through automation [2]. The efficacy of LLMs in automating fundamental pre-processing tasks streamlines initial data analysis stages [4, 11, 15, 17], enabling human analysts to focus on complex, domain-specific analytical challenges.

Planning-level User Agency. In contrast to their significant enthusiasm for assistance at the execution level, participants showed interest and caution regarding planning-level assistance. One reason for their caution is a lack of confidence in the quality of plans generated by LLMs. While using OHA, most participants opted to create execution commands rather than consult the LLM's analysis plan. The planning mechanism in SLA, which generates a step-by-step guide based on user questions and domain knowledge, demonstrated limited capabilities and may not suit complex real-world tasks. Some participants were interested in whether the LLM-powered tool could assist in brainstorming potential analytical approaches and facilitating iterative exploration. However, they emphasized that the tool should offer suggestions, while the final decision on which suggestion to adopt should rest with them as data scientists. This finding underscores the importance users place on the planning-level agency. It is imperative that design considerations reflect this priority.

4.2 Calibrating User Trust and Expectations with AI Explainability in Data Analysis Assistance

During our probe, **participants exhibited a general caution towards the results** generated by SLA, mainly due to the **absence of methods to verify these outcomes** and an expectation for **consistent results**. Additionally, they also raised questions concerning the process by which the LLM reached specific conclusions, impacting their trust in the outputs of LLM-based systems collectively. While this behavior aligned with prior work on data scientists seeking interpretable results [18], this skepticism could be rooted in the well-documented hallucination tendencies of LLMs [13]. It could be beneficial to offer users a verification sandbox playground, allowing them to manually conduct their analyses and compare results with those generated by the LLM, aiding in **calibrating expectation and understanding the capabilities of an LLM-based system**.

The quest for transparency was evident when participants, with enhanced domain knowledge, showed a preference for step-by-step elucidation of the analyses, assuming that SLA followed these steps. This hints at the potential benefit of **visualizing the analytical**

Interaction	OHA Probe	SLA Probe
Data Context Establishment	Copied sample data or columns from Jupyter notebook, e.g., "Write python code that takes a pandas dataframe with the following fields <copied_text>." Drafted prompt in a notepad then paste to chat textbox. Only described useful columns. Directly asked questions without giving any information about the data.	Selected dataset or upload own csv files. Manually typed column description in the data description text box.
Domain Knowledge Injection	Manually typed in the text box. Instead of giving a formal definition of domain concepts, directly gave concrete instructions of calculation or plotting.	Manually provided input and generated variation(s). Identified and corrected incorrect augmented domain knowledge.
Error Handling and Debugging	Copied the partial or entire error message from Jupyter notebook. Tried to debug by themselves	Modified the question or domain knowledge and run analysis again.
Output Interpretation	Read the plot or transformed data, gave explanation based on their own interpretation.	Read through the "insights" generated by LLM, critiqued and revised what is wrong.

Table 2: Summary of participants' interactions with the two LLM-based tools used in the interview study.

process through a flowchart or similar means, displaying intermediate steps, and even showcasing counterfactual results at each juncture for better user validation. Enhancing transparency — visualizing backend operations and ensuring step-by-step explainability — can demystify the process, nurturing trust and engagement with the system.

5 LIMITATIONS AND FUTURE WORK

In this study, we only interviewed data scientists with experience with manufacturing and transportation data. As some participants pointed out, the tool can be beneficial for users who have rich domain knowledge but little programming skills. Future work would propose new design ideas with the suggestions we list above. We also would like to consider different design options for users with different skills and needs.

6 CONCLUSION

We explored the design of AI-powered, domain-specific data analysis tools through two distinct prototypes, differing in interaction styles and user agency, and conducted interviews with data scientists to assess their effectiveness. Our findings showed that while users appreciated the assistance of LLMs, they also emphasized the need for transparency, result validation, and different interaction workflows catered to varying levels of agency. This research opens the door for future studies on the integration of LLMs into domain-specific applications, focusing on enhancing user collaboration, explainability, and adaptable interaction modes.

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Data Profile

Please provide some context about this database.

This dataset is about furniture companies and the products.

Select the dataframes(s) you will use in your analysis

manufacturer × furniture × furniture_manuf... ×

Dataframe: manufacturer . Shape: (7, 5)

Show all data

	Manufacturer_ID	Open_Year	Name	Num_of_Factories	Num_of_Shops
0	1	1,980	Chevrolet House	36	8
1	2	1,990	IKEA	21	19
2	3	1,991	Ford Make	12	2
3	4	1,992	Jiaju	1	35
4	5	2,000	Chevrolet	38	24

Provide your description of the dataframe for manufacturer

Data type of each column: Manufacturer_ID: int, Open_Year: float, Name: str, Num_of_Factories: int, Num_of_Shops: int

Dataframe: furniture . Shape: (7, 4)

Figure 3: Probe SLA: Adding Data Context

Designer Playground

Question

Is Scalability Indicator relevant to the manufacturer age? Does manufacturers with longer history have higher Scalability Indicator score?

Domain Knowledge. Provide more context about the question. If specific terms in the question require understanding of some domain knowledge, please give a description about the term and steps to analyze it.

Scalability Indicator is a measurement that looks at the ratio between number of factories to number of shops. A higher Scalability Indicator means a manufacturer has more ability to expand the market.

Enhance Original Domain Knowledge

Skip Formatting and Run Analysis

The "Scalability Indicator" is a calculated metric that is derived from taking the ratio of the total number of production facilities (factories) to the total number of retail outlets (shops). It serves as a measure of a manufacturing entity's ability to expand its market reach.

A higher "Scalability Indicator" implies that the manufacturer has a larger production capacity in relation to its retail footprint. This can be interpreted as the manufacturer having the potential to increase its retail outlets and expand its market, without being limited by production capacity.

This concept can be applied in the context of our data by considering tracks as "factories" (production facilities) and races as "shops" (retail outlets).

- Calculate the total number of unique tracks (Track_ID) in df1 (This represents the total number of "factories").
- Calculate the total number of unique races (Race_ID) in df0 (This represents the total number of "shops").
- Compute the "Scalability Indicator" by dividing the total number of unique tracks by the total number of unique races.

Please note that this interpretation might not directly apply if the business context doesn't match with the analogy. Always validate these derivations with the subject matter experts in your team.

Figure 4: Probe SLA: Providing Domain Knowledge and Enhancing it with the LLM

